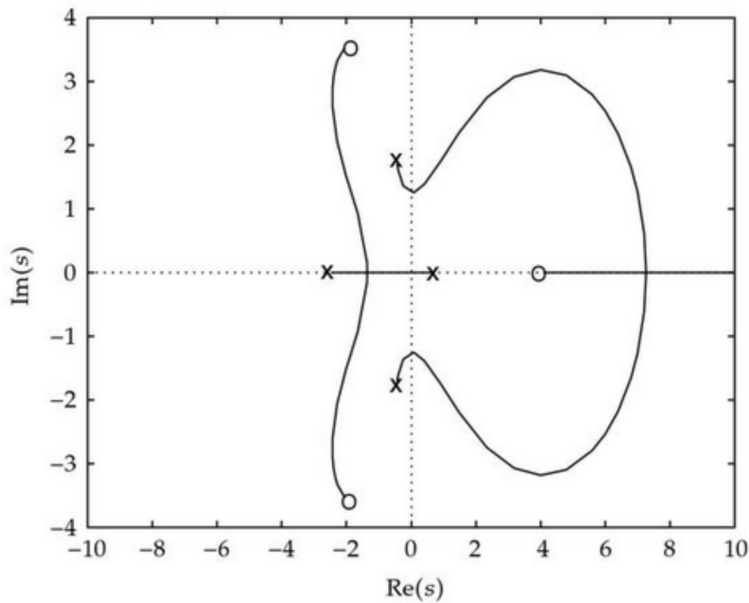


- Last time we looked at the root-locus. A plot that shows us how a closed-loop system behaves for different gains.



Poles move along a path towards the zeros.

The location of poles and zeros tells us a lot about a system!

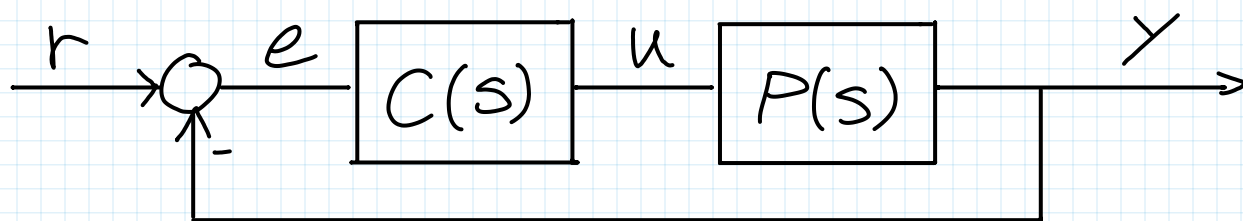
- To stabilize a system, all poles of the closed loop have to be in the left half of the complex plane.

Great, system stabilized, job done.

Well... Unfortunately just stabilizing isn't enough. We need to also make sure it's quick and handles errors and disturbances well.

But first, let's build some intuition

- Let's follow the path...

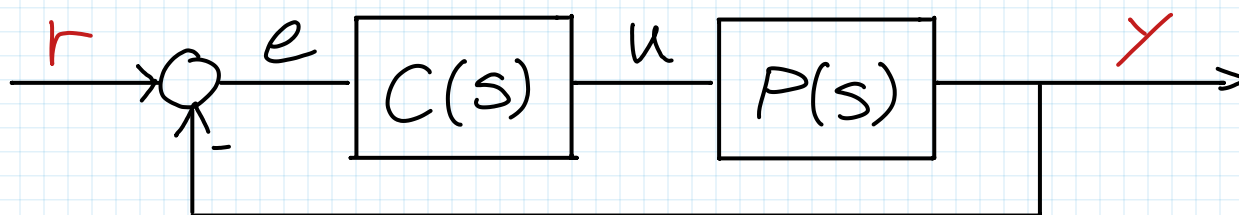


In essence, the controller calculates necessary input based on how far off the output is from the reference.

- To analyze how well this works, we need to mainly look at two different tf's:

Complementary sensitivity

- How well does the output y follow the reference r ?



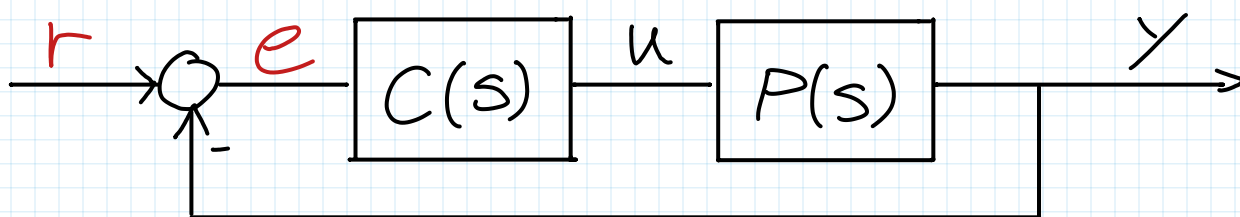
For this we must look at the tf from $r \rightarrow y$, called the complementary sensitivity:

$$T(s) = \frac{C(s)P(s)}{1 + C(s)P(s)}$$

Note: $T(s)$ is also just called closed-loop tf.

Sensitivity

- How well does the error e vanish, given a reference r ?



- For this we look at the tf from $r \rightarrow e$, called sensitivity.

$$S(s) = \frac{1}{1 + C(s)P(s)}$$

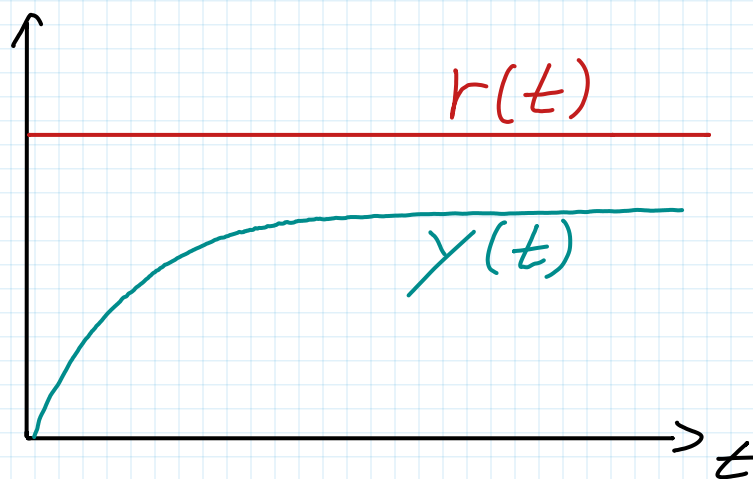
Wait, sensitivity and complementary sensitivity? What do they have to do with each other? Take a look:

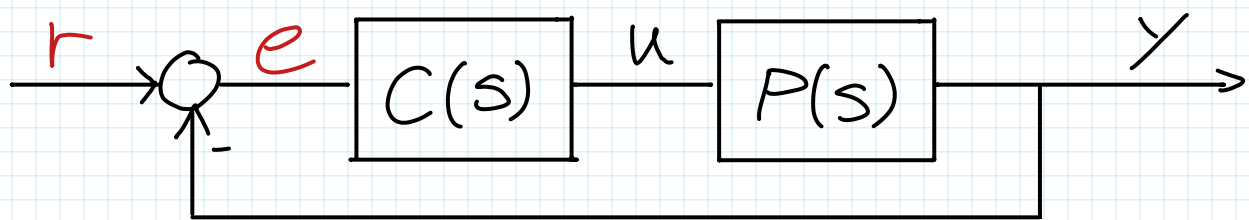
$$T(s) + S(s) = \frac{CP}{1+CP} + \frac{1}{1+CP} = 1$$

$\Rightarrow$ We will be using $T(s)$ and $S(s)$ to evaluate performances of systems.

Steady-state-error

- Assume we apply a step input to a system. Does the output approach the reference?





From this graph we know that $e = r - y$.
As previously defined, the relationship between r and e is captured with

$$\frac{E(s)}{R(s)} = S(s) = \frac{1}{1 + C(s)P(s)}$$

We find the steady-state-error as

final value theorem $\mathcal{L}\{e(t)\}$

$$e_{ss} = \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} s \cdot E(s) = \lim_{s \rightarrow 0} s \cdot S(s) \cdot R(s)$$

step input $R(s) = \frac{1}{s}$, hence

$$e_{ss} = \lim_{s \rightarrow 0} s \cdot \frac{1}{1 + C(s)P(s)} \cdot \frac{1}{s} = \frac{1}{1 + C(0)P(0)}$$

This means, that for e_{ss} to be

zero, we need $L(0) = C(0)P(0) = \infty$.

• What feature must the general $L(s)$ have for this to be true?

$\Rightarrow$ We need a pole at $s=0$!
Integrators!!

• If a constant error is integrated, it steadily grows. The controller tries to minimize this growth and drives $e(t)$ to zero.

Ex: Given an open loop system:

$$L(s) = \frac{1}{s+2}$$

What is e_{ss} for step-input?

$$e_{ss} = S(0) = \frac{1}{1+L(0)} = \frac{1}{1+\frac{1}{2}} = \frac{2}{3} = 0,6\bar{6}$$

Let's add an **integrator**

$$L(s) = \frac{1}{s} \cdot \frac{1}{s+2} = \frac{1}{s^2 + 2s}$$

$$e_{ss} = \frac{1}{1 + \frac{1}{0}} = 0$$

Matlab

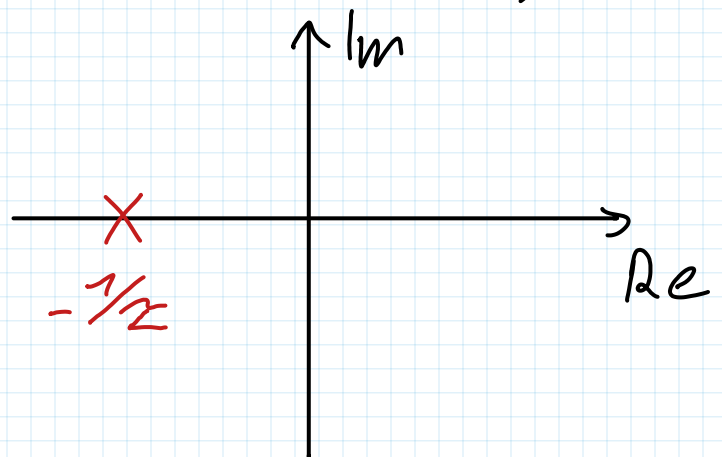
Great, now we know how to make the output actually go where we want it to go.

• Next topic to analyze: How fast is our system?

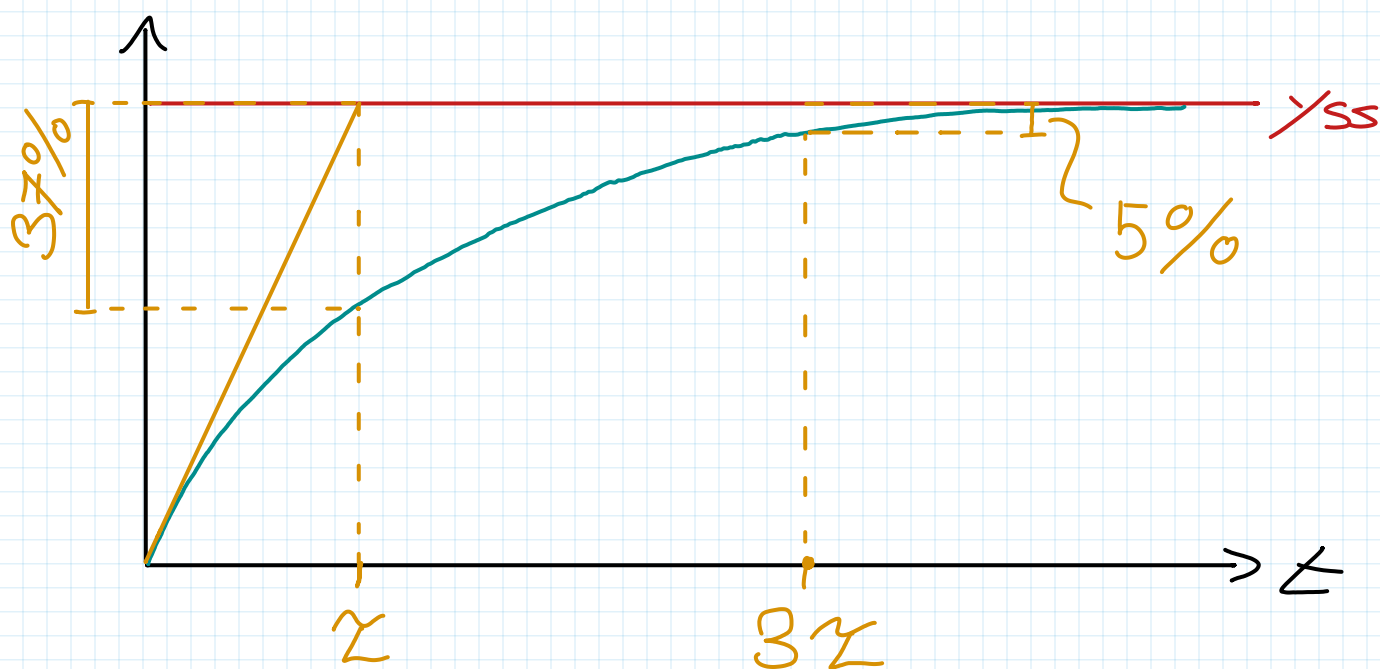
• Let's start with a first order system

$$G(s) = \frac{1}{\tau s + 1}$$

$$\tau \in \mathbb{R}$$



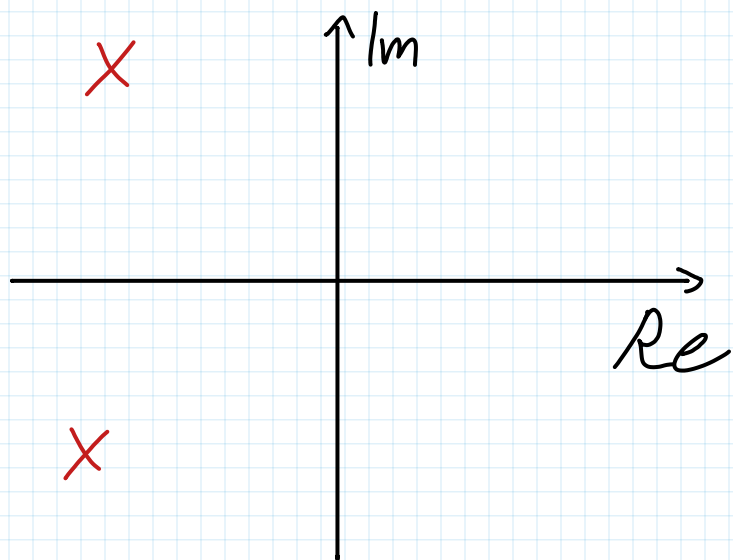
It turns out we can conveniently sketch the transient response knowing τ , the time-constant.



• Only with τ , we can plot response quite accurately!

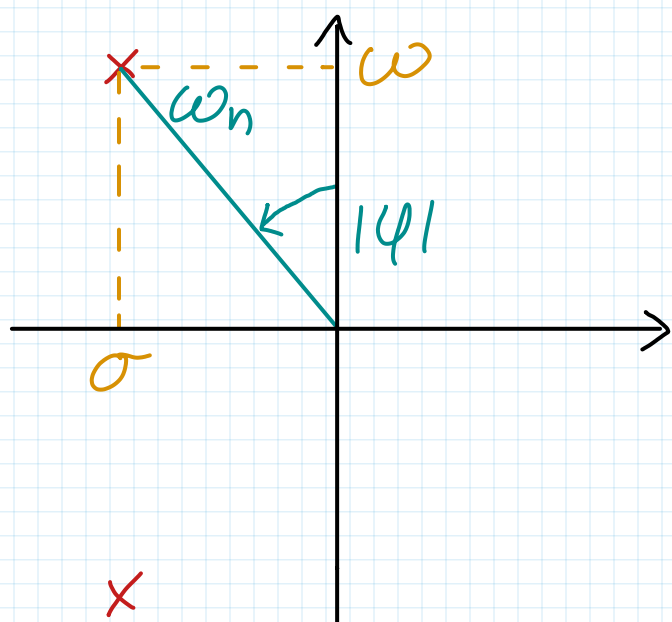
Well what about systems that aren't this simple?

• As an example:



If we know the exact location of the poles, we know everything about the transient response!

- We will have to define a couple things...

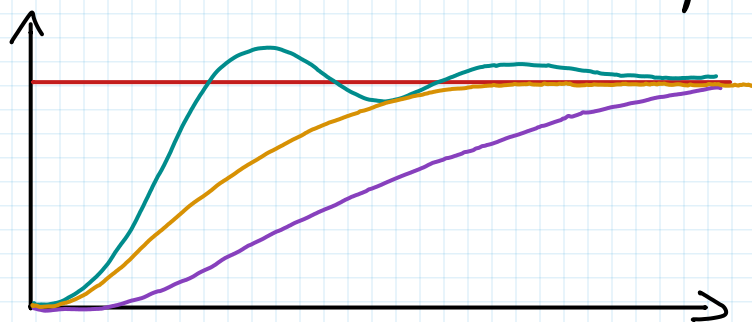


- $\omega_n = \sqrt{\sigma^2 + \omega^2}$
- $\varphi = \tan^{-1}\left(\frac{\sigma}{\omega}\right)$
- $|\sigma| = \sin(|\varphi|) \cdot \omega_n$
- $\sin(|\varphi|) = \zeta$

- Knowing these, the transfer function can be written as:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

- ζ is called damping factor.



$$0 \leq \zeta < 1$$

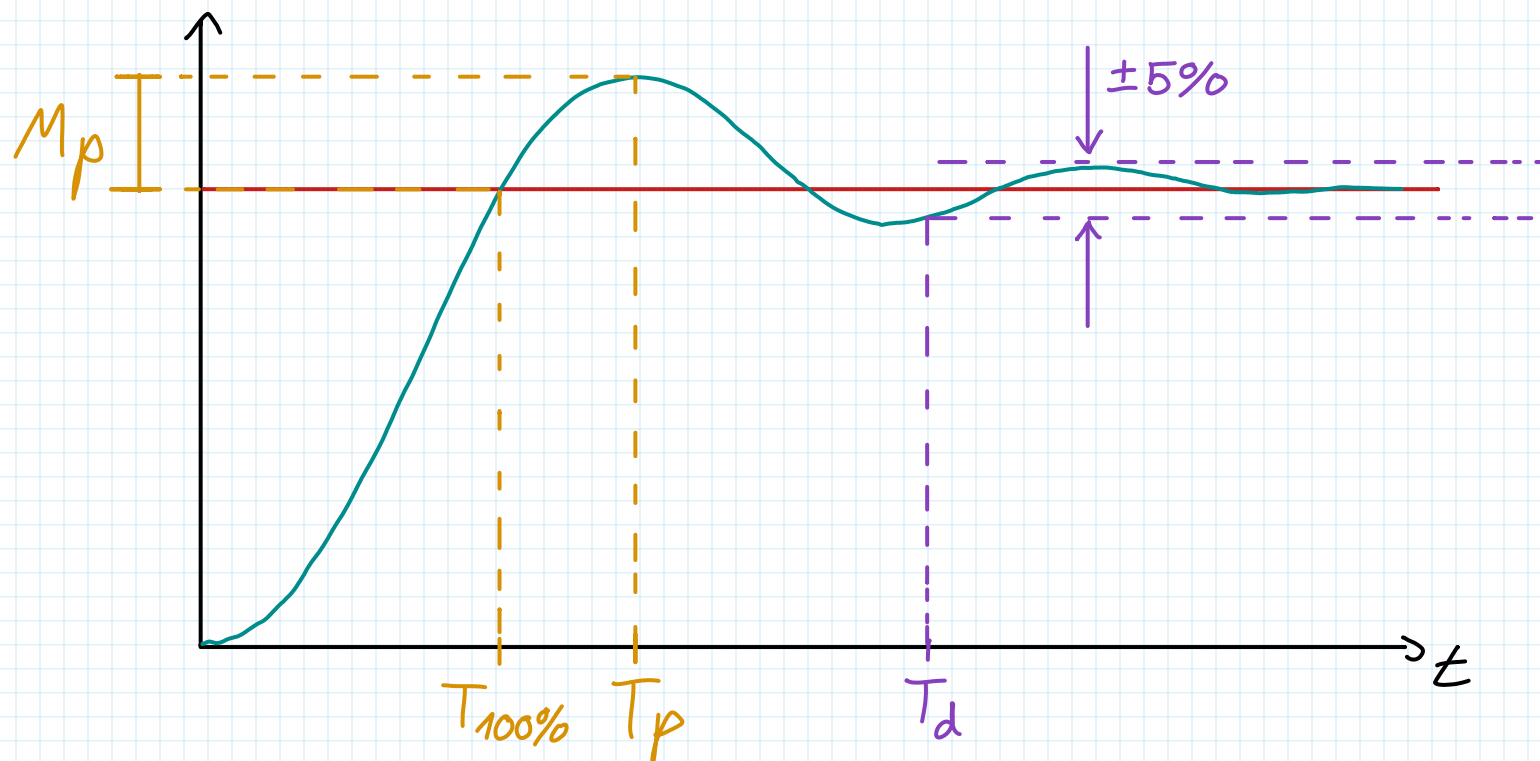
$$\zeta = 1$$

$$\zeta > 1$$

under-
critically
over-

} damped

Assuming an underdamped system, we can characterize the oscillatory response using the previously defined terms.



$$\bullet T_{100\%} = \frac{\pi/2 - \varphi}{\omega}$$

$$\bullet T_p = \frac{\pi}{\omega}$$

$$\bullet M_p = \exp\left(\frac{\sigma \pi}{\omega}\right)$$

$$\bullet T_d = \frac{1}{|\sigma|} \log(100/d)$$

$\Rightarrow$ Here $d\% = 5\%$

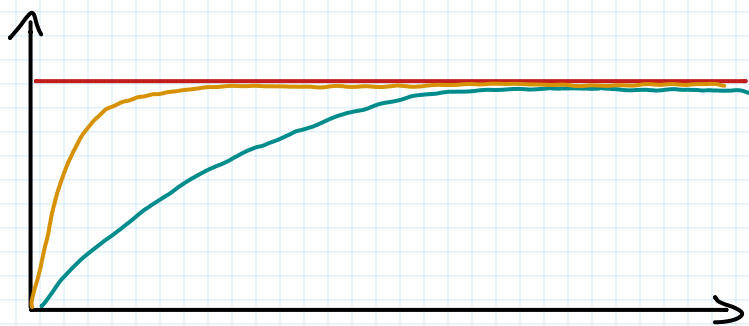
$\Rightarrow$ All of this information, just from the location of the poles!

So we can characterize 1st and 2nd order systems now. What about the all other systems?

Thought experiment:

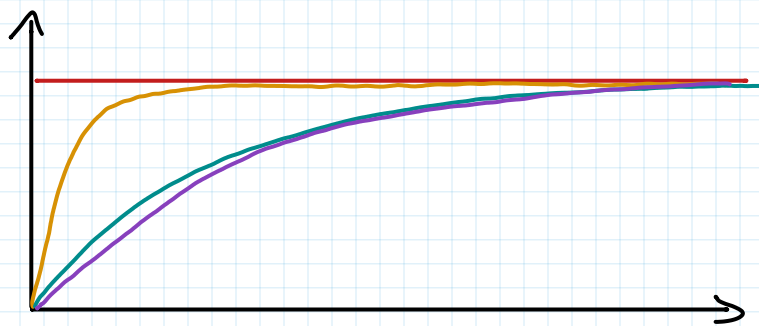
$$G_1(s) = \frac{1}{(s+1)}$$

$$G_2(s) = \frac{1}{(s+5)}$$



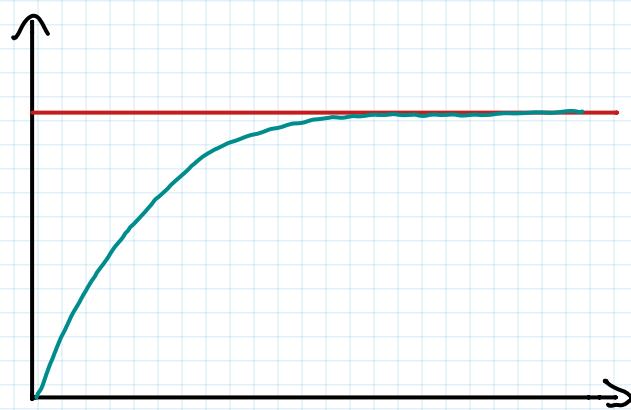
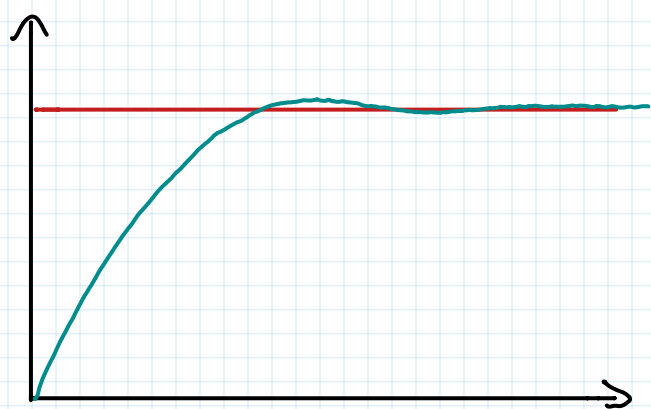
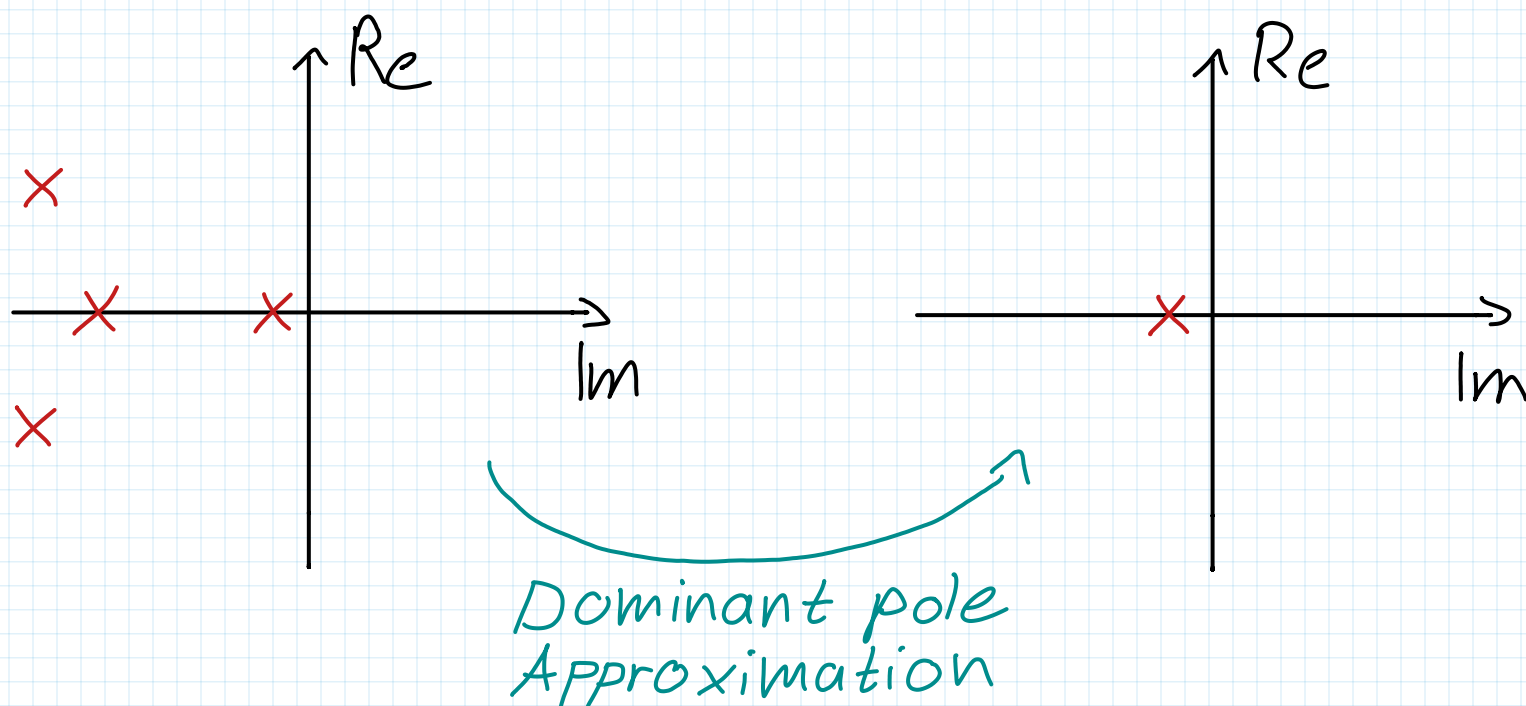
• What do you expect for a system with both poles?

$$G_3(s) = \frac{1}{(s+1)(s+5)} = \frac{1}{s^2 + 6s + 5}$$



$\Rightarrow$ Poles closer to the origin have a higher influence!!

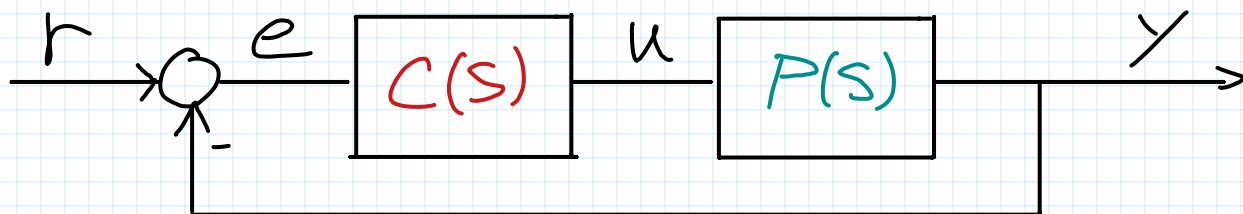
- This means, we can approximate higher order systems with a 2nd or even first order system.



Now we need to take a step back and assess how this all comes together.

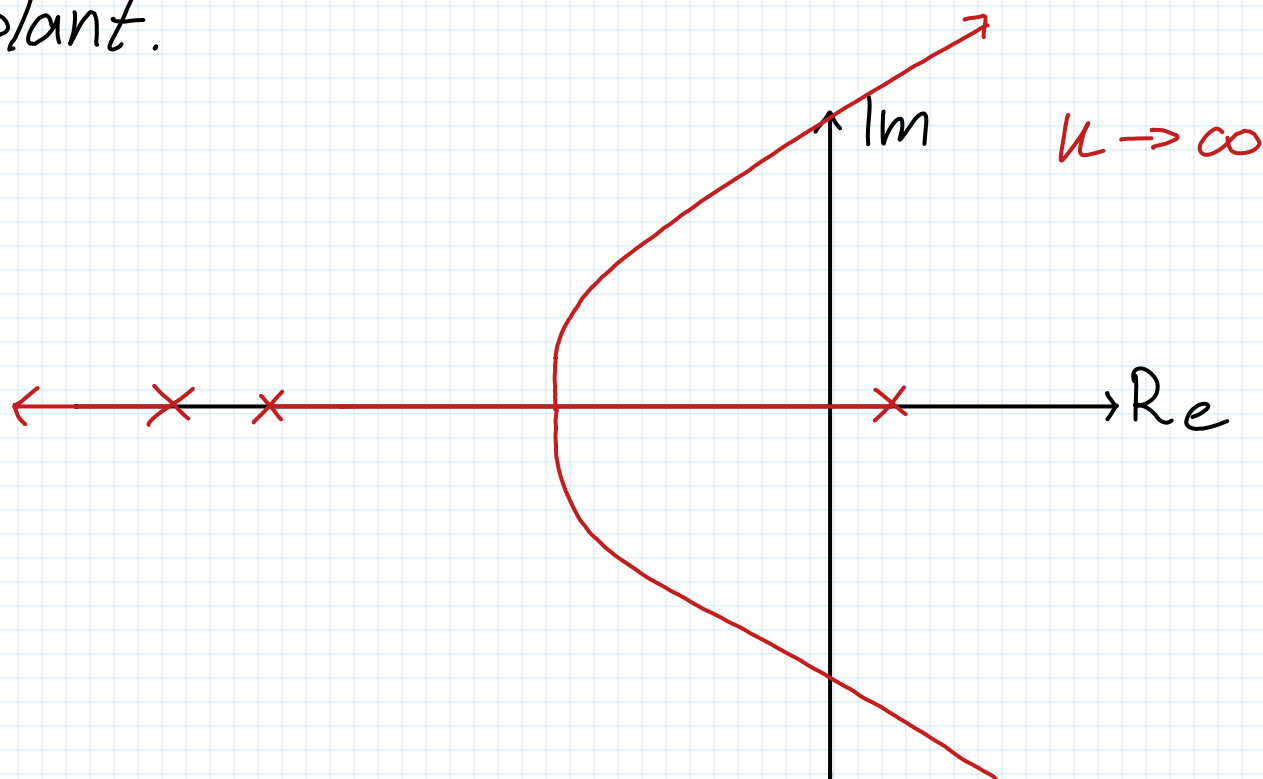
Control design process example:

- You are given a **Plant** and tasked to find a **controller**



$$P(s) = \frac{1}{(s-1)(s+3)(s+5)} = \frac{1}{s^3 + 7s^2 + 7s - 15}$$

- First you choose a proportional controller $C(s) = k$ and plot the root locus to see if you can stabilize the plant.



- The root locus suggest that stabilizing is possible, so how exactly should you choose k ?
- We analyze the step response!
Since we know exactly how to do this for 2nd order systems, we will approximate:

$$P(s) \approx P'(s) = \frac{a}{(s-1)(s+3)} = \frac{a}{s^2+2s-3}$$

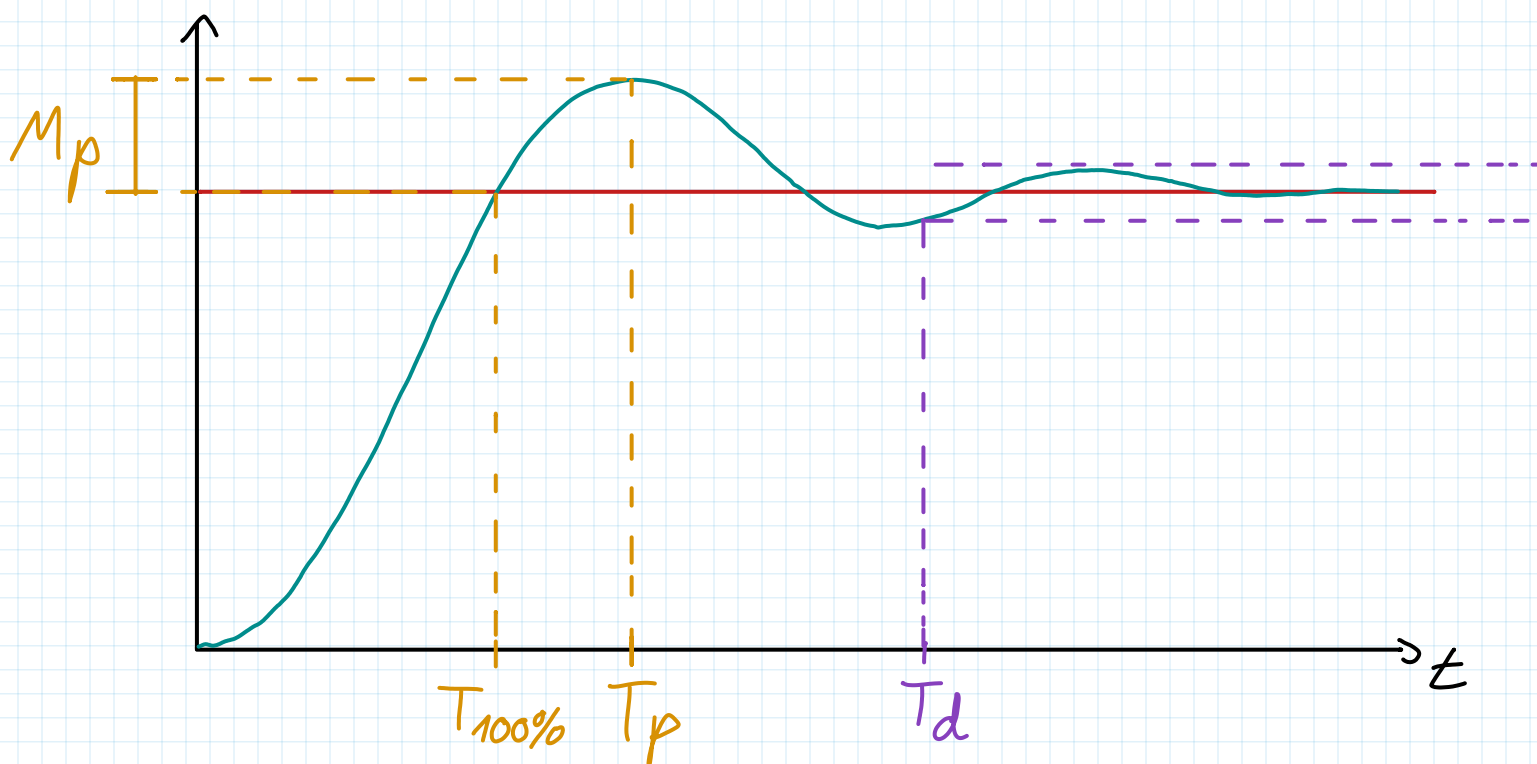
- To find a , set $P'(0) = P(0)$

$$\frac{a}{-3} = \frac{1}{-15} \Rightarrow P'(s) = \frac{1}{5s^2+10s-15}$$

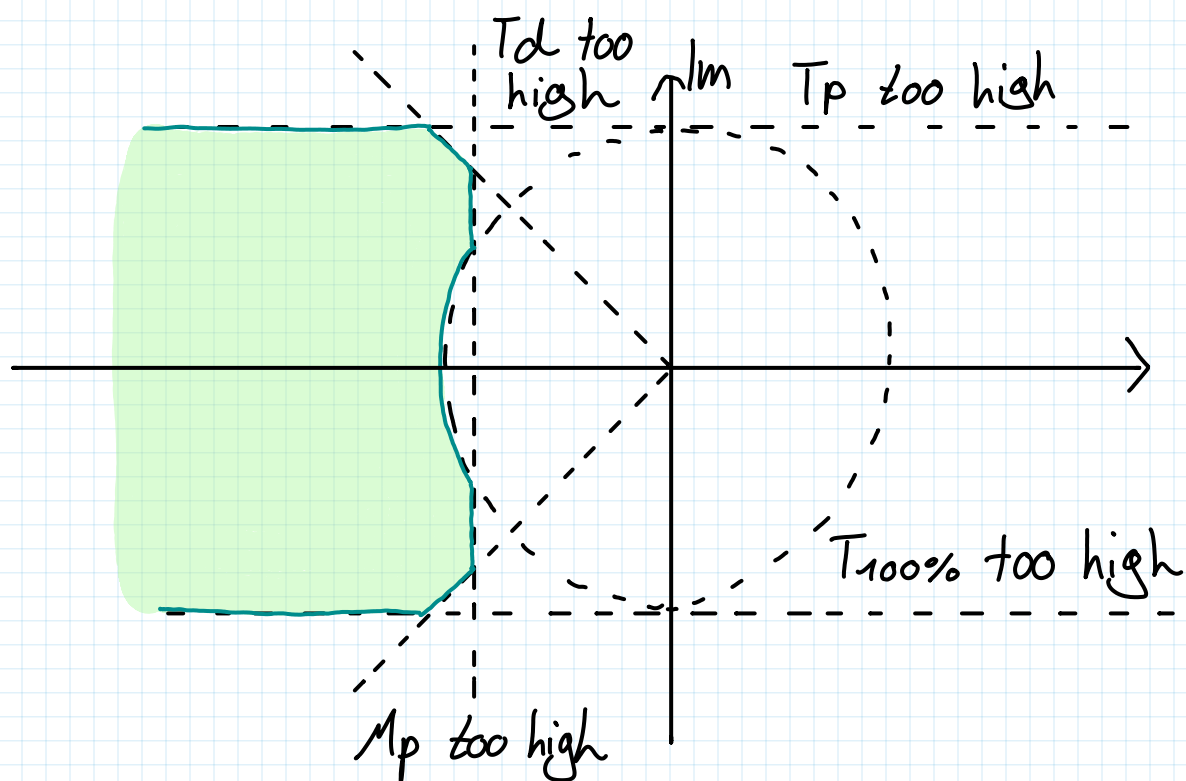
- Now we evaluate the closed loop tf

$$T(s) = \frac{k \cdot P'(s)}{1 + k \cdot P'(s)} = \frac{k}{5s^2+10s-15+k}$$

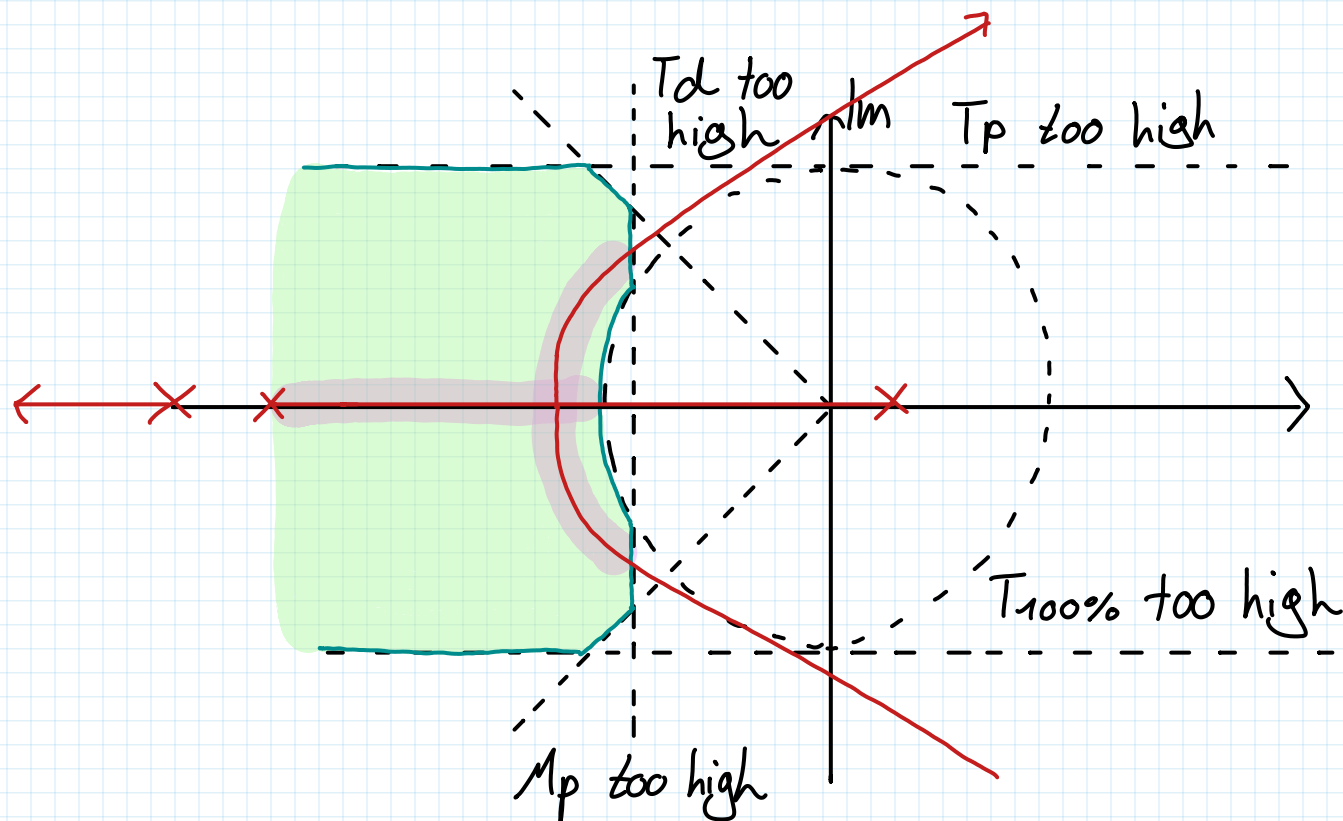
And by varying k can find a fitting step.



Here the restrictions are:



Now we just have to overlap this with the possible locations of the poles (root locus):

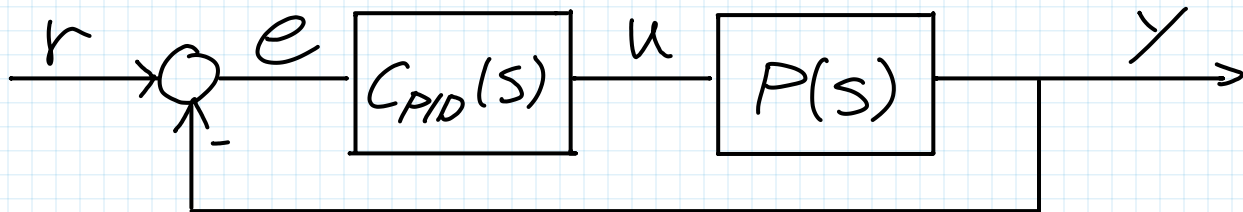


Now we choose k , such that the closed loop poles are in this area.

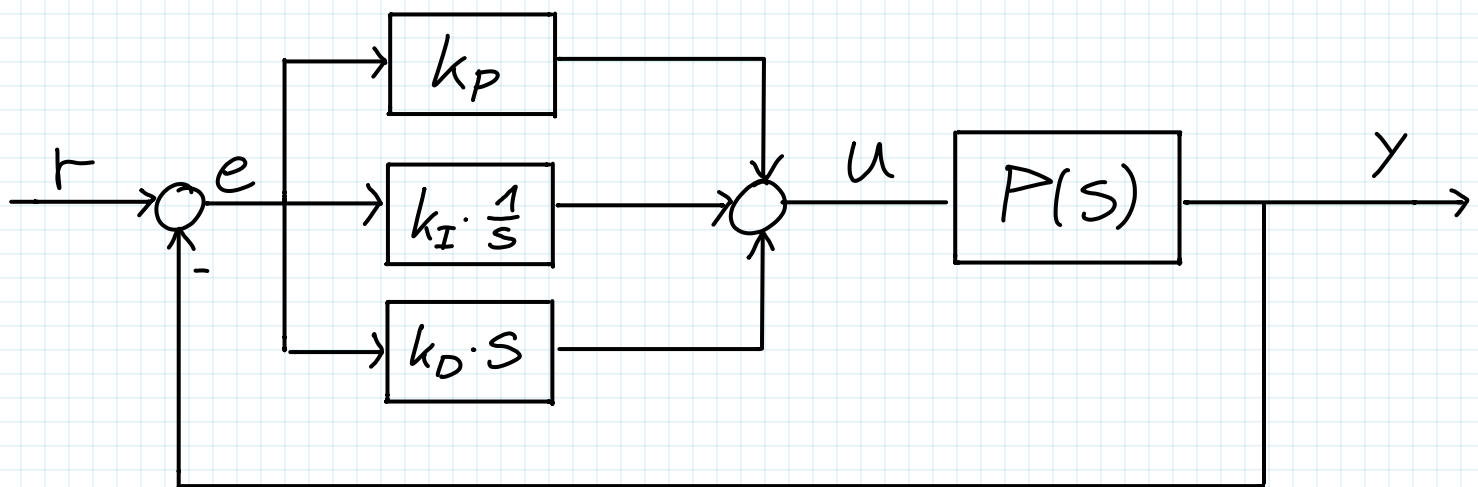
Often just proportional control isn't enough... What if choose a PID?

- General structure of a PID-controller:

$$C_{PID}(s) = k_P + k_I \cdot \frac{1}{s} + k_D \cdot s$$



- Which really is just this:



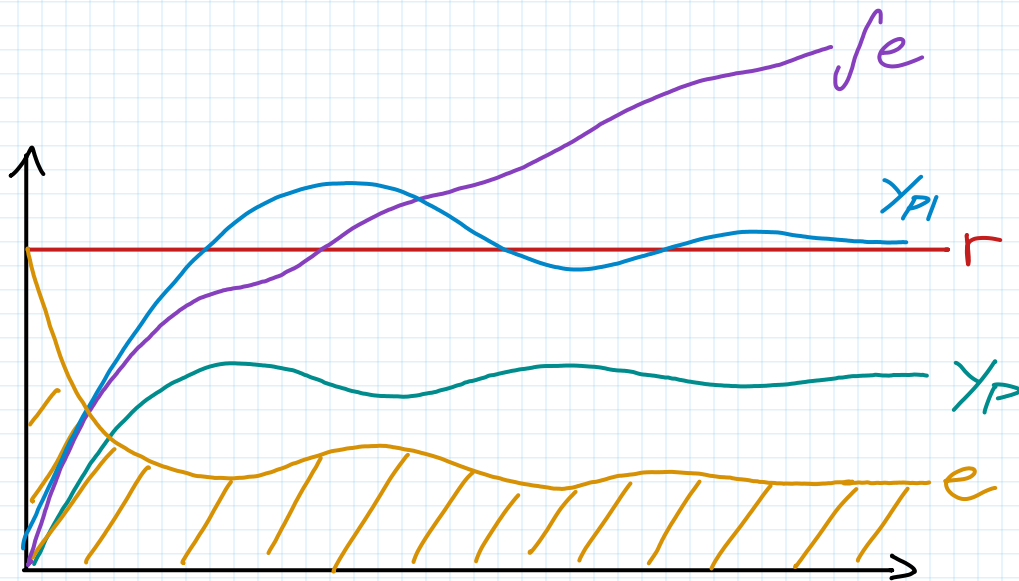
- Let's break down how the derivative and integral action works.

Integrator

- The integrator integrates the error

$$e = r - y$$

$\Rightarrow$ Punishes area between the signals



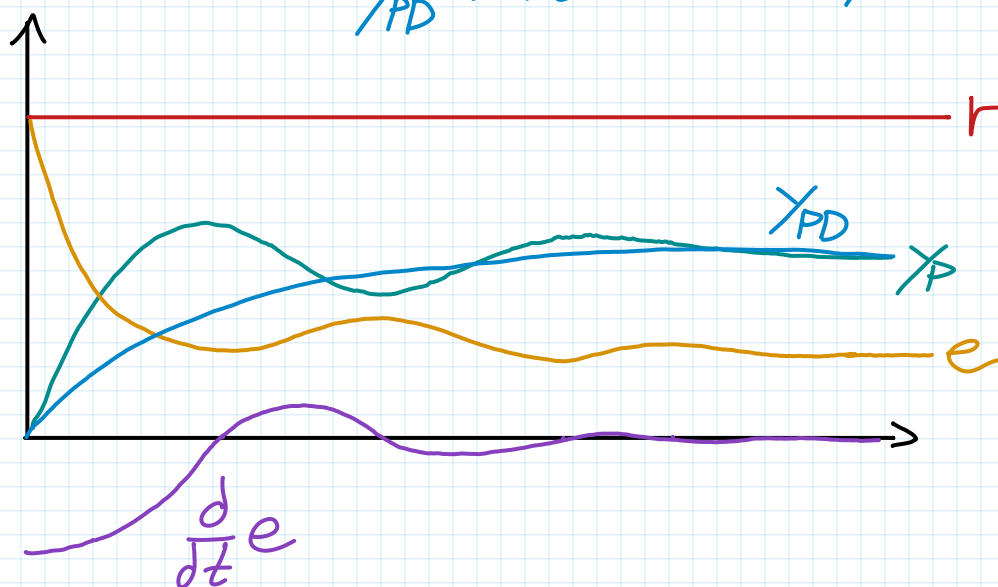
- y_P with $C = k_P$
- y_{PI} with $C = k_P + k_I \cdot \frac{1}{s}$

Differentiator

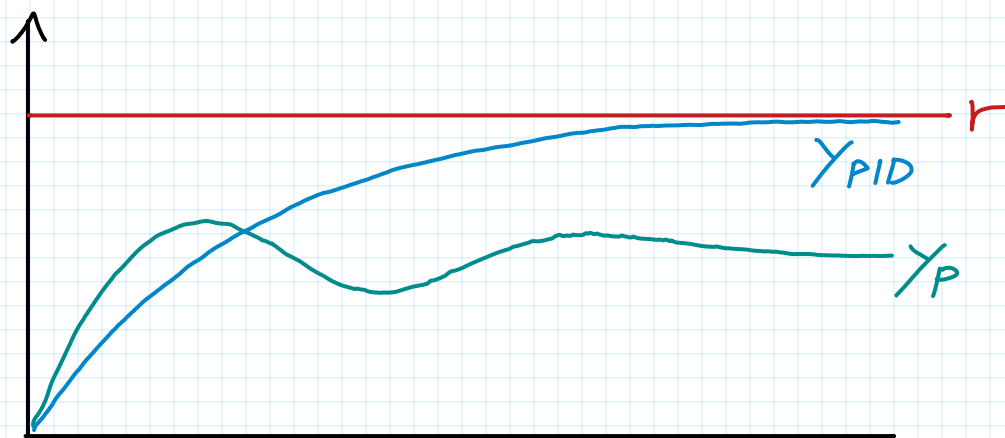
- The differentiator takes the derivative of the error $e = r - y$.

$\Rightarrow$ Punishes steep changes of error

- y_P with $C = k_P$
- y_{PD} with $C = k_P + k_D \cdot s$



Combining it all in a full PID-controller



- y_P with $C = k_P$
- y_{PID} with $C = k_P + k_I \cdot \frac{1}{s} + k_D \cdot s$
- And by adjusting k_P, k_I, k_D we choose how strong each effect is!

Q1) Given a 2nd order system of this form:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Increasing ζ will result in

☐ Faster response

☐ More oscillation

☒ Slower response

☒ Less oscillation

Q2) Find the 2nd order dominant pole approximation of:

$$P(s) = \frac{10}{(s+1+j)(s+1-j)(s+6)}, \quad P(0) = \frac{10}{2 \cdot 6} = \frac{5}{6}$$

$$P'(s) = \frac{k}{(s+1+j)(s+1-j)}, \quad P'(0) = \frac{k}{2}$$

$$P(0) = P'(0) \Rightarrow \frac{5}{6} = \frac{k}{2} \Rightarrow k = \frac{5}{3}$$

$$P'(s) = \frac{5}{3(s+1+j)(s+1-j)}$$

Q3)

Problem: Consider the standard feedback architecture shown in Figure 19 where a PID -controller $C(s) = k_P + k_D s + k_I/s$ is used to control a linear time-invariant system P . Further, assume that P can be described by a strictly proper rational transfer function with no poles or zeros at $s = 0$.

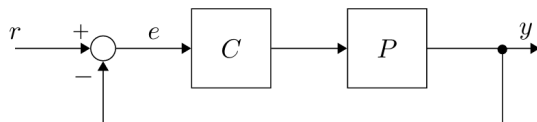


Figure 19: Standard feedback architecture. System P and controller C .

Q35 (1 Points) Mark all correct statements.

- ☒ Assume k_I and k_D fixed. Increasing k_P can make the system respond faster.
- ☒ Assume k_D and k_P fixed. In order to guarantee a zero steady-state error $e_{ss} = 0$ to step references, it is necessary that $k_I \neq 0$.
- ☐ Assume k_I and k_P fixed. By tuning k_D we can reduce the steady-state error e_{ss} to step references to zero.
- ☒ Assume k_I and k_P fixed. By increasing k_D we can reduce the overshoot of the closed-loop step response.